Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Engineering · Class 12

Subject: Physics

Medium: English

Academic Year 2026-27

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This book is prepared as a teaching guide for teachers.

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Unit 1: Electromagnetic Induction: Faraday's Law

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Learning Objectives

• Students will state Faraday's law and explain magnetic flux in their own words.

• Students will use Lenz's law to predict the direction of an induced current.

• Students will explain why generators and transformers depend on changing flux.

Key Concepts

• Magnetic flux is the number of magnetic field lines passing through a loop.

• Faraday's law says a changing magnetic flux induces a voltage in a coil.

• Lenz's law says the induced current always opposes the change that produced it.

• More turns, a stronger magnet, or faster motion all increase the induced voltage.

Teaching Methodology (Step by Step)

1. Start with the magnet-and-coil demo and ask why the needle moves only while the magnet moves.

2. Define magnetic flux with a simple loop diagram on the board.

3. State Faraday's law and link each factor to the demo: speed, strength, turns.

4. Introduce Lenz's law as nature resisting change, with a worked direction example.

5. Students predict the current direction for a magnet moving away from the coil.

6. Connect the lesson to a bicycle dynamo and a power-station generator.

Classroom Activities

• Coil and magnet demo: students predict what happens to the needle when the magnet moves faster.

• Flux sketch: groups draw field lines through a loop for three magnet positions.

• Direction game: use Lenz's law to predict the induced current direction for a new case.

Assessment Questions

1. State Faraday's law in one sentence.

2. A magnet moves towards a coil. In which direction does the induced current oppose the change?

3. Why does a faster-moving magnet produce a larger induced voltage?

Homework

• Explain with a diagram why a generator needs a rotating coil, not a still one.

• List three devices at home or school that use electromagnetic induction.

Time Allocation

40 minutes